Home Work on Histogram/Filtering/ Morphology

Submitted By

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Question: Suppose that a digital image is subjected to histogram equalization. Show that a second pass of histogram equalization will produce exactly the same result as the first pass.

Answer:

Considering an image having the following histogram:

Gray level (r_k)	0	1	2	3	4	5	6	7
No. of pixels	790	1023	850	656	329	245	122	81

Histogram equalization (Pass 1):

Gray level (r_k)	No. of pixels (n_k)	(PDF) r_k $P_r(r_k)=\frac{n_k}{N}$	CDF	$(L-1)CDF = 7\times CDF$	Rounding off	No. of pixels (n_k) o/p image
0	790	0.19	0.19	1.33	1	0
1	1023	0.25	0.44	3.08	3	790
2	850	0.21	0.65	4.55	5	0
3	656	0.16	0.81	5.67	6	1023
4	329	0.08	0.89	6.23	6	0
5	245	0.06	0.95	6.65	7	850
6	122	0.03	0.98	6.86	7	985
7	81	0.02	1	7	7	448
	$N=4096$	$\Sigma PDF = 1$				

Histogram Equalization (Pass-2):

Gray Level (r_K)	No. of pixels (n_K)	(PDF) r_K $P_r(r_K) = \frac{n_K}{N}$	CDF	(L-1).CDF = 7×CDF	Rounding off	No. of pixels (n_K) o/p image
0	0	0	0	0	0	0
1	790	0.19	0.19	1.33	1	790
2	0	0	0.19	1.33	1	0
3	1023	0.25	0.44	3.08	3	1023
4	0	0	0.44	3.08	3	0
5	850	0.21	0.65	4.55	5	850
6	985	0.24	0.89	6.23	6	985
7	448	0.11	1	7.00	7	448

So, the output image remains unchanged after 2nd pass of histogram equalization.

Question: What is the smallest number of different structuring elements that you would need to use to locate all foreground points in an image which have at least one foreground neighbor, using the hit-and miss transform? What do the structuring elements look like?

Answer: Total 4 different structuring elements needed to use to locate all foreground points in an image which have atleast one foreground neighbor using hit and miss transform. The structuring elements are the following:

1)

X	X	X
X	1	1
X	X	X

2)

X	X	X
1	1	X
X	X	X

3)

X	1	X
X	1	X
X	X	X

4)

X	X	X
X	1	X
X	1	X

Here, X → Don't care
1 → Foreground

Question:
Opening the foreground pixels with a particular structuring element is equivalent to closing the background pixels with the same element. Make an example with simulations.

Answer: Consider the following image as input:

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	1	0
0	0	1	0	0	0
0	0	0	0	0	0

Here,
1 → Foreground
0 → Background

Structuring Element

	1	
1	1	1
	1	

Now for opening have to perform Erosion first with structuring element.

After performing Erosion,

0	0	0	0	0	0
0	0	0	0	0	0
0	0	1	1	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0

After performing Dilation,

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0
0	0	0	0	0	0

Now perform complement of the original image to get the background image.

$(Input)^c$

1	1	1	1	1	1
1	1	0	0	1	1
1	0	0	0	0	1
1	1	0	0	0	1
1	1	0	1	1	1
1	1	1	1	1	1

Structuring Element.

	1	
1	1	1
	1	

We are using the same structuring element.
For closing, after applying Dialation:

1	1	1	1	1	1
1	1	1	1	1	1
1	1	0	0	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1

After applying Erosion:

1	1	1	1	1	1
1	1	0	0	1	1
1	0	0	0	0	1
1	1	0	0	1	1
1	1	1	1	1	1
1	1	1	1	1	1

Performing complement on the resulting image:

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0
0	0	0	0	0	0

This same image is also obtained after opening.
So, opening the foreground pixels with a particular SE is equivalent to closing the background pixels with the same SE.